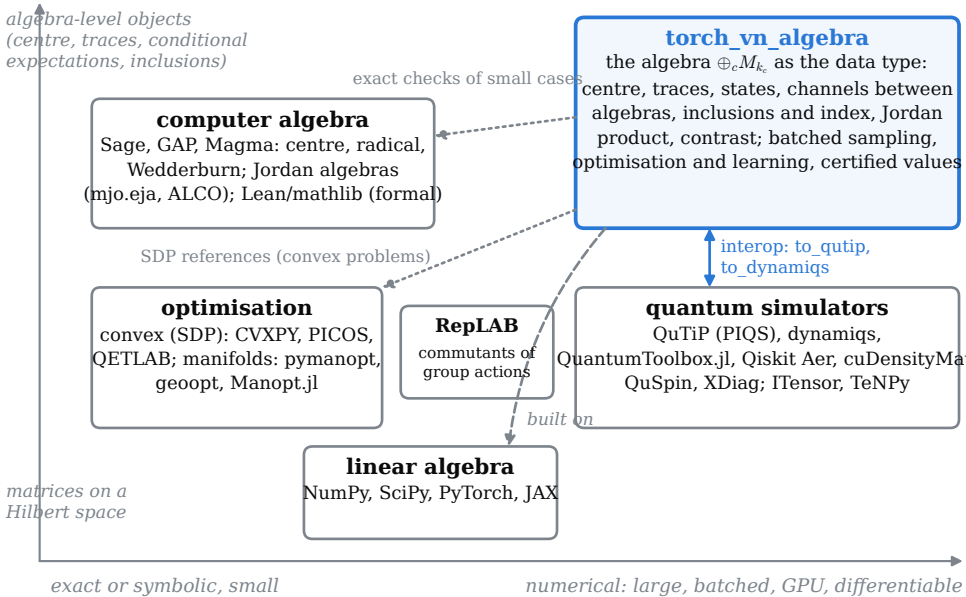
algebra-level objects
(centre, traces, conditional
expectations, inclusions)
exact checks of small cases
torch_vn_algebra
the algebra $\oplus_c M_{k_c}$ as the data type:
centre, traces, states, channels between
algebras, inclusions and index, Jordan
product, contrast; batched sampling,
optimisation and learning, certified values
computer algebra
Sage, GAP, Magma: centre, radical,
Wedderburn; Jordan algebras
(mjo.eja, ALCO); Lean/mathlib (formal)
SDP references (convex problems)
interop: to_qutip,
to_dynamiqs
optimisation
convex (SDP): CVXPY, PICOS,
QETLAB; manifolds: pymanopt,
geoopt, Manopt.jl
RepLAB
commutants of
group actions
quantum simulators
QuTiP (PIQS), dynamiqs,
QuantumToolbox.jl, Qiskit Aer, cuDensityMat;
QuSpin, XDiag; ITensor, TeNPy
built on
linear algebra
NumPy, SciPy, PyTorch, JAX
matrices on a
Hilbert space
exact or symbolic, small
numerical: large, batched, GPU, differentiable